

WEEK 5 – Healthcare Data Transformation & Data Modeling

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1. INTRODUCTION

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The **healthcare_dataset.csv** file was imported and analyzed using Power BI.

The healthcare records were first processed through **Power Query**, where the data was cleaned and transformed. A new **Age Group** field was also generated using the Conditional Column option. After the transformation process, the information was divided into appropriate tables to build an organized data model.

Relationships with **Many-to-One (*:1) cardinality** were then created between the relevant tables.

Through this activity, I developed practical understanding of data cleaning, Power Query transformations, Conditional Columns, data modeling, and table relationships in Power BI.

2. OBJECTIVE

The primary aim of this activity is to clean, organize, and structure the healthcare dataset in Power BI for further analysis.

Objectives:

- To perform data cleaning and modification using Power Query.
- To implement suitable transformation operations on the dataset.
- To generate an **Age Group** field through a Conditional Column.
- To divide the healthcare information into well-organized tables.
- To define **Many-to-One (*:1)** relationships between connected tables.
- To develop a properly structured model that can support further analysis.

3. DATASET

The analysis was carried out using the **healthcare_dataset.csv** file in Microsoft Power BI.

The dataset includes several types of healthcare and patient-related details, including:

- Patient ID
- Name
- Age
- Gender
- Blood Type
- Medical Condition
- Date of Admission
- Doctor
- Hospital
- Insurance Provider
- Billing Amount
- Room Number
- Admission Type
- Discharge Date
- Medication
- Test Results
- Admission ID

This dataset served as the source for performing data preparation, transformation, and modeling activities in Power BI.

Table.AddColumn("Cleaned Text1", "Age_group", each if [Age] <= 12 then "Child" else if [Age] <= 19 then "Teenager" else if [Age] <= 39						
Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor
1 Bobby Jackson	30	MALE	B-	Cancer	31-01-2024	Matthew Smit
2 Leslie Terry	62	MALE	A+	Obesity	20-08-2019	Samantha Da
3 Danny Smith	76	FEMALE	A-	Obesity	22-09-2022	Tiffany Mitch
4 Andrew Watts	28	FEMALE	O+	Diabetes	18-11-2020	Kevin Wells
5 Adrienne Bell	43	FEMALE	AB+	Cancer	19-09-2022	Kathleen Han
6 Emily Johnson	36	MALE	A+	Asthma	20-12-2023	Taylor Newto
7 Edward Edwards	21	FEMALE	AB-	Diabetes	03-11-2020	Kelly Olson
8 Christina Martinez	20	FEMALE	A+	Cancer	28-12-2021	Suzanne Thor
9 Jasmine Aguilar	82	MALE	AB+	Asthma	01-07-2020	Daniel Fergus
10 Christopher Berg	58	FEMALE	AB-	Cancer	23-05-2021	Heather Day
11 Michelle Daniels	72	MALE	O+	Cancer	19-04-2020	John Duncan
12 Aaron Martinez	38	FEMALE	A-	Hypertension	13-08-2023	Douglas Mayr
13 Connor Hansen	75	FEMALE	A+	Diabetes	12-12-2019	Kenneth Fletc
14 Robert Bauer	68	FEMALE	AB+	Asthma	22-05-2020	Theresa Freei
15 Brooke Brady	44	FEMALE	AB+	Cancer	08-10-2020	Roberta Stew
16 Ms. Natalie Gamble	46	FEMALE	AB-	Obesity	01-01-2023	Maria Doughv
17 Haley Perkins	63	FEMALE	A+	Arthritis	23-06-2020	Erica Spencer
18 Mrs. Jamie Campbell	38	MALE	AB-	Obesity	08-03-2020	Justin Kim
19 Luke Burgess	34	FEMALE	A-	Hypertension	04-03-2021	Justin Moore
20 Daniel Schmidt	63	MALE	B+	Asthma	15-11-2021	Denise Gallov
21 Timothy Burns	67	FEMALE	A-	Asthma	28-06-2023	Krista Smith
22 Christopher Bright	48	MALE	B+	Asthma	21-01-2020	Gregory Smit
23 Kathryn Stewart	58	FEMALE	O+	Arthritis	12-05-2022	Vanessa New

4. POWER QUERY DATA TRANSFORMATION & AGE GROUP

Before developing the data model, the healthcare records were prepared using **Power Query Editor**. Various transformation operations were applied to improve the consistency and usability of the dataset.

4.1 Data Transformation Steps

1. Promoted Headers

The first row of the dataset was set as the column header so that each field could be properly recognized.

2. Changed Type

Appropriate data types were assigned to the columns based on their contents, including text, numeric, and date values.

3. Capitalized Each Word

Required text fields were formatted using capitalization to maintain a uniform appearance throughout the dataset.

4. Removed Errors

Records containing error values were removed to improve the reliability and quality of the data.

5. Trimmed Text

Extra spaces present in text fields were eliminated using the Trim operation.

6. Cleaned Text

Unwanted and non-printable characters were removed from the relevant text fields.

7. Uppercased Text

Selected text values were converted into uppercase to maintain consistent formatting.

8. Added Index

An Index Column was introduced to assign sequential numbers to the records and assist with data organization.

9. Removed Columns

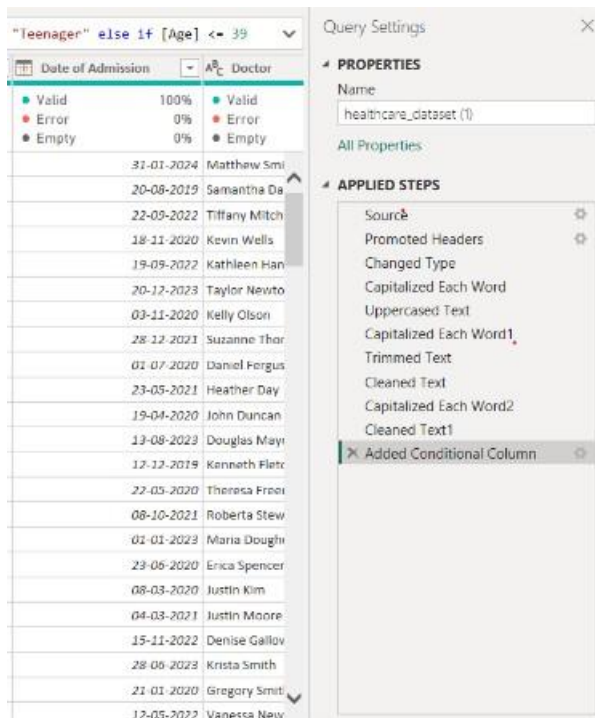
Fields that were unnecessary for the analysis were deleted from the dataset.

10. Renamed Columns

Selected field names were modified to make them clearer and easier to understand.

11. Removed Duplicates

Duplicate entries were identified and removed to maintain data consistency.



4.1 AGE GROUP USING CONDITIONAL COLUMN

A new field called **Age Group** was created with the Conditional Column feature available in Power Query.

The existing Age field was used to categorize patients into different age-based groups.

Age Condition	Age Group
Age $\leq$ 12	Child
Age $\leq$ 19	Teenager
Age $\leq$ 39	Younger Adult
Age $\leq$ 59	Elder Adult
Age > 59	Senior Adult

The age classification provides a convenient way to organize and analyze patients according to their respective age ranges.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Age group

	Column Name	Operator	Value ①		Output ②	
If	Age	is less than or equ...	ABC 123 12	Then	ABC 123 Child	...
Else If	Age	is less than or equ...	ABC 123 19	Then	ABC 123 Teenager	
Else If	Age	is less than or equ...	ABC 123 39	Then	ABC 123 Younger Adult	
Else If	Age	is less than or equ...	ABC 123 59	Then	ABC 123 Elder Adult	

Add Clause

Else ③

ABC 123

Senior Adult

OK

Cancel

5. DATA MODELING

Once the required transformations were completed, the healthcare information was arranged into a structured data model in Power BI.

The data was separated into the following tables:

Dim_Patient

This table stores the main details associated with patients.

- Patient ID
- Name
- Age
- Age Group
- Gender
- Blood Type

Dim_Admission

This table contains information related to patient admissions and medical details.

- Admission ID
- Admission Type
- Doctor
- Hospital
- Medical Condition
- Medication
- Room Number
- Test Results

Billing

This table holds billing and admission-related fields.

- Patient ID
- Admission ID
- Billing Amount
- Date of Admission
- Discharge Date
- Insurance Provider

Dividing the information into separate tables provides a more organized structure and makes the data easier to use for further analysis.

Queries [4] < ✕ ✓ fx = Table.AddColumn("#Cleaned Text1", "Age group"

	t	123 Room Number	A ^B C Admission Type
	100% 0% 0%	Valid Error Empty	100% 0% 0%
1	18856.28131	328	Urgent
2	33643.32729	265	Emergency
3	27955.09608	205	Emergency
4	37909.78241	450	Elective
5	14238.31781	458	Urgent
6	48145.11095	389	Urgent
7	19580.87234	389	Emergency
8	45820.46272	277	Emergency
9	50119.22279	316	Elective
10	19784.63106	249	Elective
11	12576.79561	394	Urgent
12	7999.58688	288	Urgent
13	43282.28336	134	Emergency
14	33207.70663	309	Urgent

healthcare_dataset (1)

- Dim_Patient
- Dim_Admission
- Billing

6. RELATIONSHIPS

After preparing the individual tables, relationships were created by using the common key fields available between them.

Relationship 1 – Patient

Billing → Dim_Patient

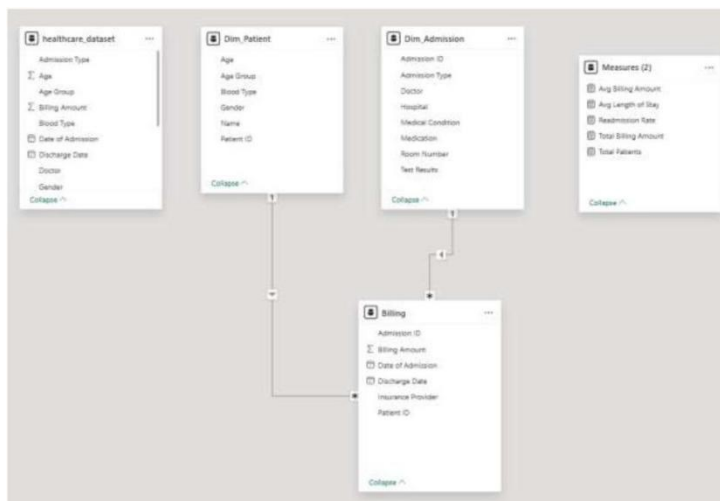
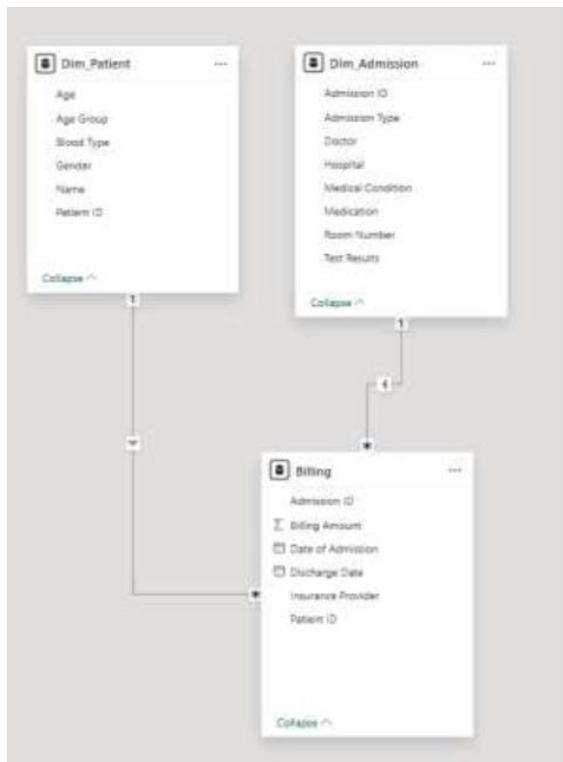
- **Common Key:** Patient ID
- **Cardinality:** Many-to-One (*:1)

Relationship 2 – Admission

Billing → Dim_Admission

- **Common Key:** Admission ID
- **Cardinality:** Many-to-One (*:1)

These connections link the related tables and make it possible to work with information from different tables together during analysis.



7. CONCLUSION

The healthcare dataset was prepared and transformed using **Power Query**, and an **Age Group** field was generated through a Conditional Column.

The processed information was then structured into **Dim_Patient**, **Dim_Admission**, and **Billing** tables. Appropriate **Many-to-One (*:1)** relationships were established using Patient ID and Admission ID as the connecting fields.